

# Probability Theory

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An overview of probability using measures. We will denote probability measures defined over  $\sigma$ -algebras with  $\mathbb{P}$  and probability functions defined over some sample space  $\Omega$  or  $\mathbb{R}$  with  $P$  or  $p$ . When we say countable, we mean finite or countably infinite. I have used resources from:

1. Rick Durrett's *Elementary Probability and Probability* textbooks.
2. Dr. Krishna's *Probability Foundation for Electrical Engineers* lectures at IIT.
3. Various quant interview books and websites for examples.

## 1 Jan 12

### Example 1.1 (Branching Processes)

Let  $Z_n$  be the number of people in the  $n$ th generation, and  $X_i^{(n+1)}$  be the number of offspring the  $i$ th person in the  $n$ th generation has. Assume  $X_i^{(n+1)}$  is iid with distribution  $X$ . Then, we have

$$Z_{n+1} = X_1^{(n+1)} + X_2^{(n+1)} + \dots + X_{Z_n}^{(n+1)}, \quad Z_1 = 1 \quad (1)$$

What is the extinction probability  $\pi$  for given  $X$ ?

$$\pi = \mathbb{P}(\exists n \text{ s.t. } Z_n = 0) \quad (2)$$

Some ideas is that we can try to compute  $\mathbb{E}[X]$ , use a Markov chain. We can also use a generating function, which is defined

$$f(\theta) = \mathbb{E}[\theta^X] = \sum_{k=0}^{\infty} \theta^k \mathbb{P}(X = k) \quad (3)$$

Then, we can define  $f_n(\theta) = \mathbb{E}[\theta^{Z_n}]$ . Let  $\pi_n = \mathbb{P}(Z_n = 0)$ . By the dominated convergence theorem,

$$\pi = \lim_{n \rightarrow \infty} \pi_n \quad (4)$$

We claim that  $f_n(\theta) = f_{n-1}(f(\theta))$ , which can be proven by the Tower property:

$$f_n(\theta) = \mathbb{E}[\mathbb{E}[\theta^{Z_n} \mid Z_{n-1}]] \quad (5)$$

$$= \mathbb{E}[\mathbb{E}[\theta^{X_1^{(n)} + \dots + X_{Z_{n-1}}^{(n)}} \mid Z_{n-1}]] \quad (6)$$

$$= \mathbb{E}[f(\theta)^{Z_{n-1}}] \quad (7)$$

$$= f_{n-1}(f(\theta)) \quad (8)$$

where the final line is true since  $f_{n-1}(\alpha) = \mathbb{E}[\alpha^{Z_{n-1}}]$ . This gives us a nice recursive formula, which implies that

$$\pi_n = f_n(0) = f_{n-1}(f(0)) \quad (9)$$

$$= \underbrace{(f \circ \dots \circ f)}_{n \text{ times}}(0) \quad (10)$$

$$= f(f_{n-1}(0)) \quad (11)$$

$$= f(\pi_{n-1}) \quad (12)$$

So  $\pi$  satisfies  $\pi = f(\pi)$  as long as  $f$  is continuous.

So let's actually do some computation.

1. Assume  $X \sim \text{Geom}(p)$  i.e.  $\mathbb{P}(X = k) = (1-p)^k p$  for  $0 < p < 1$ . Then,

$$f(\theta) = \frac{p}{1 - (1-p)\theta} \quad (13)$$

Since  $\theta = f(\theta)$ , we can solve

$$\theta = \frac{p}{1 - (1-p)\theta} \implies \theta = 1 \text{ or } \theta = \frac{p}{1-p} \quad (14)$$

So how do we know which solution is  $p$ ? If  $p \geq 1/2$ , then  $\pi = 1$ , but if  $\frac{p}{p-1} < 1$ , then we can derive further

$$f_n(0) = \frac{p\left(\frac{1-p}{p}\right) - p}{(1-p)\left(\frac{1-p}{p}\right)^n - p} \rightarrow \frac{p}{1-p} \quad (15)$$

as  $n \rightarrow \infty$ , so  $\pi = \frac{p}{1-p}$  is the correct extinction probability.

It turns out that according to some deeper theorem, if  $\mathbb{E}[X] > 1$ , then  $\pi$  is the unique solution to  $x = f(x)$  on  $(0, 1)$ , and if  $\mathbb{E}[X] \leq 1$ , then  $\pi = 1$ .

### Example 1.2

We can describe  $Z_n$  more precisely as  $n \rightarrow \infty$  using a Markov chain, which can be stated in the equivalent ways.

1.

$$\mathbb{P}(Z_{n+1} = j \mid Z_0 = i_0, \dots, Z_n = i_n) = \mathbb{P}(Z_{n+1} = j \mid Z_n = i_n) \quad (16)$$

2.

$$\mathbb{E}[Z_{n+1} \mid Z_0, \dots, Z_n] = \mathbb{E}[Z_{n+1} \mid Z_n] \quad (17)$$

Observe that  $M_n = Z_n/\mu^n$  is a martingale, with  $\mu = \mathbb{E}[X]$ , i.e.  $\mathbb{E}[M_n \mid M_{n-1}] = M_{n-1}$ . So,

$$\mathbb{E}[M_n \mid M_{n-1}] = \mathbb{E}[Z_n \mid Z_{n-1}] = \mu \frac{Z_{n-1}}{\mu_n} = \frac{Z_{n-1}}{\mu^{n-1}} = M_{n-1} \quad (18)$$

By a Martingale convergence theorem, setting  $M_\infty(\omega) = \lim_{n \rightarrow \infty} M_n(\omega)$ , we have  $M_n \rightarrow M_\infty$  a.s.

Next time, we'll review conditional expectation, then martingales, then markov chains, then ergodic theory (building on convergence results), and finally with Brownian motion.

## 2 Jan 14

### Definition 2.1 (Conditional Expectation)

Given probability space  $(\Omega, \mathcal{F}_0, \mathbb{P})$ , a  $\sigma$ -algebra  $\mathcal{F} \subseteq \mathcal{F}_0$  and a random variable  $X$  with  $\mathbb{E}[X] < +\infty$ , we define the **conditional expectation** of  $X$  given  $\mathcal{F}$ —denoted  $\mathbb{E}[X \mid \mathcal{F}]$ —as *any* random variable (called a **version**)  $Y : \Omega \rightarrow \mathbb{R}$  s.t.

1.  $Y$  is  $\mathcal{F}$ -measurable.
2. For all  $A \in \mathcal{F}$ ,  $\int_A X d\mathbb{P} = \int_A Y d\mathbb{P}$

*Proof.* We should claim that it always exists and it is unique up to measure 0.

1. *Uniqueness* Show that if  $Y$  is a version, then  $\mathbb{E}[|Y|] < +\infty$ . Hint: Show that  $\mathbb{E}[|Y|] < \mathbb{E}[|X|]$ . We split  $Y = Y_+ - Y_-$ . Then,  $\mathbb{E}[|Y|] = \mathbb{E}[Y_+] + \mathbb{E}[Y_-]$ , and we can bound

$$\mathbb{E}[Y_+] = \int_{\mathbb{R}_{>0}} Y_+ d\mathbb{P} = \int_{\mathbb{R}_{>0}} X d\mathbb{P} \leq \int_{\mathbb{R}_{\geq 0}} |X| d\mathbb{P} \quad (19)$$

$$\mathbb{E}[Y_-] = \int_{\mathbb{R}_{<0}} Y_- d\mathbb{P} = \int_{\mathbb{R}_{<0}} X d\mathbb{P} \leq \int_{\mathbb{R}_{<0}} |X| d\mathbb{P} \quad (20)$$

Combining these together, we get  $\mathbb{E}[|Y|] < \mathbb{E}[|X|]$ . To prove uniqueness, we can see that if  $Y, Y'$  are two versions of  $\mathbb{E}[X | \mathcal{F}]$ , then let  $A_\epsilon = \{\omega \in \Omega: Y(\omega) - Y'(\omega) > \epsilon\}$ . Then,

$$\mathbb{P}(A)\epsilon \leq \int_A (Y - Y') d\mathbb{P} = \int_A (X - X) = 0 \implies Y \leq Y' \text{ a.s.} \quad (21)$$

Now swap  $Y$  and  $Y'$  to get  $Y' \leq Y$ .

2. *Existence.* To prove existence of  $Y$ , this is much more involved. Recall the definition of absolutely continuous measures and  $\sigma$ -finite measures, along with the Radon-Nikodym theorem. We will use this to construct our conditional expectation. Since we can write  $X = X_+ - X_-$ , WLOG let  $X \geq 0$ . Let  $\mu = \mathbb{P}$ , we are looking for  $Y$  s.t. for all  $A \in \mathcal{F} \subset \mathcal{F}_0$ ,

$$\int_A Y d\mu = \int_A X d\mu \quad (22)$$

Now set  $\nu(A) := \int_A X d\mu$ , which turns out to be a  $\sigma$ -finite measure on  $(\Omega, \mathcal{F})$ . By construction,  $\nu \ll \mu$ . By the Radon-Nikodym theorem,  $\exists f$  s.t. for all  $A \in \mathcal{F}$ ,

$$\int_A f d\mu = \nu(A) = \int_A X d\mu \quad (23)$$

Now set  $f = Y$ .

So it is the best approximation in a sense. The motivation is that it is a projection of  $X$  onto the space of  $\mathcal{F}$ -measurable random variables.

### Example 2.1

Suppose  $X, Z$  are random variables with a joint pdf  $f_{X,Z}(x, z)$  and  $\mathbb{E}[|X|] < +\infty$ . Then, we define the conditional pdf as

$$f_{X|Z}(x | z) := \begin{cases} \frac{f_{X,Z}(x, z)}{f_Z(z)} & \text{if } f_Z(z) \neq 0 \\ 0 & \text{if } f_Z(z) = 0 \end{cases} \quad (24)$$

Define  $Y(z) := \int_{\mathbb{R}} x f_{X|Z}(x | z) dx$ . We claim that  $Y(Z)$  is a version of  $\mathbb{E}[X | \sigma(Z)]$ . As a proof, for all  $A \in \sigma(Z)$ , we have

$$\int_A Y(z) f_Z(z) dz = \int_A \left( \int_{\mathbb{R}} x f_{X|Z}(x | z) dx \right) f_Z(z) dz \quad (25)$$

$$= \int_A \int_{\mathbb{R}} x f_{X,Z}(x, z) dx dz \quad (26)$$

where we denote  $dx, dz$  are the Lebesgue or counting measure, depending on whether it is discrete or continuous.

Now we list all of the properties of conditional expectation.

### Theorem 2.1 (Tower Rule)

Let  $X$  be a random variable on  $(\Omega, \mathcal{F}, \mathbb{P})$  with  $\mathbb{E}[|X|] < +\infty$ . Let  $\mathcal{G}, \mathcal{H}$  be sub- $\sigma$ -algebras of  $\mathcal{F}$ . If  $Y$  is any version of  $\mathbb{E}[X | \mathcal{G}]$ , then  $\mathbb{E}[X] = \mathbb{E}[Y]$ .

*Proof.*  $\Omega \in \mathcal{G}$  implies that

$$\mathbb{E}[Y] = \int_{\Omega} Y d\mathbb{P} = \int_{\Omega} X d\mathbb{P} = \mathbb{E}[X] \quad (27)$$

### Theorem 2.2

If  $X$  is  $\mathcal{G}$ -measurable, then  $\mathbb{E}[X | \mathcal{G}] = X$  a.s.

*Proof.*  $X$  is a version of  $\mathbb{E}[X | \mathcal{G}]$  because  $X$  is  $\mathcal{G}$ -measurable.

### Theorem 2.3 (Linearity)

We have

$$\mathbb{E}[aX_1 + bX_2 | \mathcal{G}] = a\mathbb{E}[X_1 | \mathcal{G}] + b\mathbb{E}[X_2 | \mathcal{G}] \quad (28)$$

*Proof.* We use linearity of the integral. For all  $A \in \mathcal{G}$ ,

$$\int_A (aX_1 + bX_2) d\mathbb{P} = a \int_A X_1 d\mathbb{P} + b \int_A X_2 d\mathbb{P} = a \int_A \mathbb{E}[X_1 | \mathcal{G}] d\mathbb{P} + b \int_A \mathbb{E}[X_2 | \mathcal{G}] d\mathbb{P} \quad (29)$$

### Theorem 2.4 (Positivity)

If  $X \geq 0$ , then  $\mathbb{E}[X | \mathcal{G}] \geq 0$  a.s.

*Proof.* Let  $Y$  be a version of  $\mathbb{E}[X | \mathcal{G}]$ . If  $\mathbb{P}(Y < 0) > 0$ , then there exists a  $\epsilon > 0$  s.t.  $\mathbb{P}(Y < -\epsilon) > 0$ . Set  $A = \{\omega \in \Omega | Y(\omega) < -\epsilon\}$ . Then,

$$-\epsilon\mathbb{P}(A) > \int_A Y d\mathbb{P} = \int_A X d\mathbb{P} \geq 0 \quad (30)$$

a contradiction.

### Theorem 2.5 (Conditional Monotone Convergence Theorem)

If  $0 \leq X_n$  is a monotonically increasing sequence of nonnegative functions and  $X_n \rightarrow X$  a.s., then

$$\mathbb{E}[X_n | \mathcal{G}] \nearrow \mathbb{E}[X | \mathcal{G}] \text{ a.s.} \quad (31)$$

*Proof.* Let  $Y_n$  be a version of  $\mathbb{E}[X_n | \mathcal{G}]$ . Set  $Y(\omega) := \limsup_n Y_n(\omega)$ , which is  $\mathcal{G}$ -measurable and  $Y_n \nearrow Y$ . It remains to show that  $Y$  is a version of  $\mathbb{E}[X | \mathcal{G}]$ . For all  $A \in \mathcal{G}$ , by MCT and the fact that  $Y_n \geq 0$ , we have

$$\int_A Y_n d\mathbb{P} \rightarrow \int_A Y d\mathbb{P} \quad (32)$$

But since  $X_n \nearrow X$ ,  $\int_A Y_n d\mathbb{P} = \int_A X_n d\mathbb{P} \rightarrow \int_A X d\mathbb{P}$ , so  $\int_A Y d\mathbb{P} = \int_A X d\mathbb{P}$ , which implies that  $Y$  is a version of  $\mathbb{E}[X | \mathcal{G}]$ .

**Theorem 2.6 (Conditional Fatou's Lemma)**

If  $X_n \geq 0$ , then

$$\mathbb{E}[\liminf_n X_n \mid \mathcal{G}] \leq \liminf_n \mathbb{E}[X_n \mid \mathcal{G}] \text{ a.s.} \quad (33)$$

**Theorem 2.7 (Conditional Dominated Convergence Theorem)**

If  $|X_n| \leq V$  for all  $n \in \mathbb{N}$  and  $\mathbb{E}[|V|] < +\infty$ , and  $X_n \rightarrow X$  a.s., then  $\mathbb{E}[X_n \mid \mathcal{G}] \rightarrow \mathbb{E}[X \mid \mathcal{G}]$  a.s.

**Theorem 2.8 (Conditional Jensen's Inequality)**

If  $f: \mathbb{R} \rightarrow \mathbb{R}$  is convex, and  $\mathbb{E}[|f(X)|] < +\infty$ , then

$$f(\mathbb{E}[X \mid \mathcal{G}]) \leq \mathbb{E}[f(X) \mid \mathcal{G}] \quad (34)$$

As a corollary, we have  $\|\mathbb{E}[X \mid \mathcal{G}]\|_p \leq \|X\|_p$  for  $p \geq 1$ , so conditional expectation is a contraction.

*Proof.* use the fact that  $f(x) = \sup_n (a_n x + b_n)$  for some sequence  $\{(a_n, b_n)\}_n$ , and then invoke linearity.

**Theorem 2.9 (Tower Rule)**

If  $\mathcal{H}$  is a sub- $\sigma$ -algebra of  $\mathcal{G}$ , then

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{H}] = \mathbb{E}[X \mid \mathcal{H}] \quad (35)$$

So, the smaller  $\sigma$ -algebra always wins.

**Theorem 2.10 (Taking Out What's Known)**

If  $Z$  is  $\mathcal{G}$ -measurable and bounded, then

$$\mathbb{E}[ZX \mid \mathcal{G}] = Z\mathbb{E}[X \mid \mathcal{G}] \quad (36)$$

*Proof.* Start out with indicators, then simple, then bounded as pointwise limits of simple functions.

**Theorem 2.11 (Role of Independence)**

If  $\mathcal{H}$  is independent of  $\sigma(\mathcal{G}, \mathcal{H})$ , then

$$\mathbb{E}[X \mid \sigma(\mathcal{G}, \mathcal{H})] = \mathbb{E}[X \mid \mathcal{G}] \text{ a.s.} \quad (37)$$

In particular,  $\mathbb{E}[X \mid \mathcal{H}] = \mathbb{E}[X]$  a.s.

*Proof.* Use  $\pi$ -systems and such, in Durrett's book.

### 3 Jan 21

Remember that for a random variable  $X$  and  $\sigma$ -algebras  $\mathcal{F}_1, \mathcal{F}_2$  s.t.  $\mathcal{F}_1 \subset \mathcal{F}_2$ , by the tower property,

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{F}_2] \mid \mathcal{F}_1] = \mathbb{E}[X \mid \mathcal{F}_1] \quad (38)$$

If we exchange the two, the smaller one wins. but for arbitrary  $\sigma$ -algebras, equality does not hold.<sup>1</sup>

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{F}_2] \mid \mathcal{F}_1] \neq \mathbb{E}[\mathbb{E}[X \mid \mathcal{F}_1] \mid \mathcal{F}_2] \quad (39)$$

It is useful to consider when  $\mathcal{F}_2 \not\subset \mathcal{F}_1$  and  $\mathcal{F}_1 \not\subset \mathcal{F}_2$ .

#### Example 3.1

The prototypical example is  $\Omega = \{a, b, c\}$  with  $\mathbb{P}(\{a\}) = \mathbb{P}(\{b\}) = \mathbb{P}(\{c\}) = \frac{1}{3}$ . Then,  $X(a) = 1, X(b) = 2, X(c) = 3$ . Consider the following  $\sigma$ -algebras:

1.  $\mathcal{F}_1$  generated by  $\{a, b\}, \{c\}$ .
2.  $\mathcal{F}_2$  generated by  $\{a\}, \{b, c\}$ .

Then,

$$\mathbb{E}[X \mid \mathcal{F}_1](\omega) = \begin{cases} 1.5 & \text{if } \omega \in \{a, b\} \\ 3 & \text{if } \omega \in \{c\} \end{cases} \implies \mathbb{E}[\mathbb{E}[X \mid \mathcal{F}_1] \mid \mathcal{F}_2] = \begin{cases} 1.5 & \text{if } \omega \in \{a\} \\ 2.75 & \text{if } \omega \in \{b, c\} \end{cases} \quad (40)$$

$$\mathbb{E}[X \mid \mathcal{F}_2](\omega) = \begin{cases} 1 & \text{if } \omega \in \{a\} \\ 2.5 & \text{if } \omega \in \{b, c\} \end{cases} \implies \mathbb{E}[\mathbb{E}[X \mid \mathcal{F}_2] \mid \mathcal{F}_1] = \begin{cases} 1.75 & \text{if } \omega \in \{a, b\} \\ 2.5 & \text{if } \omega \in \{c\} \end{cases} \quad (41)$$

#### Definition 3.1 (Filtration)

A collection of increasing  $\sigma$ -algebras  $\mathcal{F} = (\mathcal{F}_n)_{n=0}^\infty$  with  $\mathcal{F}_n \subset \mathcal{F}_{n+1}$  is a **filtration**. A sequence of random variables  $(X_n)_{n=0}^\infty$  is **adapted** to  $\mathcal{F}$  if for every  $n$ ,  $X_n$  is  $\mathcal{F}_n$ -measurable.

#### Definition 3.2 (Martingale)

If  $X_n$  further satisfies

1.  $\mathbb{E}[|X_n|] < +\infty$  (so that we don't run into any problems in cord exp)
2.  $\mathbb{E}[X_n \mid \mathcal{F}_{n-1}] = X_{n-1}$ .

Then  $(X_n)$  is called a **martingale**. If instead of (2), we have

1.  $\mathbb{E}[X_n \mid \mathcal{F}_{n-1}] \leq X_{n-1}$ , then  $(X_n)$  is a **supermartingale**.
2.  $\mathbb{E}[X_n \mid \mathcal{F}_{n-1}] \geq X_{n-1}$ , then  $(X_n)$  is a **submartingale**.

#### Example 3.2 (Random Walk)

Let  $Y_1, \dots$  be iid random variables and define

$$S_n = S_0 + \sum_{i=1}^n Y_i \quad (42)$$

where  $S_0$  is a constant. To specify a filtration, a canonical way of constructing it is to just let it be generated by the random variables. So,

$$\mathcal{F}_0 = \{\emptyset, \Omega\}, \quad \mathcal{F}_n = \sigma(Y_1, \dots, Y_n) \quad (43)$$

<sup>1</sup>No direct connection to Fubini's theorem.



For (1) to hold, just assume  $\mathbb{E}[|Y_i|] < +\infty \implies \mathbb{E}[|S_n|] < +\infty$  by triangle inequality. So, by our construction, we get

$$\mathbb{E}[S_n | \mathcal{F}_{n-1}] = S_0 + \underbrace{\sum_{i=1}^{n-1} Y_i}_{=S_{n-1}} + \underbrace{\mathbb{E}[Y_n | \mathcal{F}_{n-1}]}_{=\mathbb{E}[Y_n]} \quad (44)$$

where  $\mathbb{E}[Y_n | \mathcal{F}_{n-1}] = \mathbb{E}[Y_n]$  since  $Y_n$  is independent of  $\mathcal{F}_{n-1}$ . So,

1. if  $\mathbb{E}[Y_n] = 0$ ,  $S_n$  is a martingale.
2. if  $\mathbb{E}[Y_n] > 0$ ,  $S_n$  is a submartingale.
3. if  $\mathbb{E}[Y_n] < 0$ ,  $S_n$  is a supermartingale.

### Example 3.3

Now let

$$M_n = \prod_{i=1}^n Y_i, \quad Y_i \geq 0 \quad (45)$$

and assume  $\mathbb{E}[Y_i]^2 < +\infty$ . Then, by Cauchy-Schwarz,  $\mathbb{E}[|M_n|] < +\infty$ . Then,

$$\mathbb{E}[M_n | \mathcal{F}_{n-1}] = \left( \prod_{i=1}^{n-1} Y_i \right) \mathbb{E}[Y_n] = M_{n-1} \mathbb{E}[Y_n] \quad (46)$$

So,

1. if  $\mathbb{E}[Y_n] = 1$ ,  $S_n$  is a martingale.
2. if  $\mathbb{E}[Y_n] > 1$ ,  $S_n$  is a submartingale.
3. if  $\mathbb{E}[Y_n] < 1$ ,  $S_n$  is a supermartingale.

Now for some abstract properties of martingales.

### Lemma 3.1

If  $X_n$  is a supermartingale, then for all  $n > m$ ,  $\mathbb{E}[X_n | \mathcal{F}_m] \leq X_m$ .

*Proof.* Induct on  $n$  and tower property.

$$\begin{aligned} \mathbb{E}[X_n | \mathcal{F}_m] &= \mathbb{E}[\mathbb{E}[X_n | \mathcal{F}_{n-1}] | \mathcal{F}_m] && \text{(Tower Property)} \\ &\leq \mathbb{E}[X_{n-1} | \mathcal{F}_m] && \text{(By Definition)} \\ &\leq X_m && \text{(Inductive Hypothesis)} \end{aligned}$$

### Corollary 3.2

List.

1. If  $X_n$  is a submartingale, then for all  $n > m$ ,  $\mathbb{E}[X_n | \mathcal{F}_m] \geq X_m$ .
2. If  $X_n$  is a martingale, then for all  $n > m$ ,  $\mathbb{E}[X_n | \mathcal{F}_m] = X_m$ .

*Proof.* We use the common “symmetry trick.”

1.  $-X_n$  is a supermartingale and use the lemma.
2.  $X_n$  is both a supermartingale and submartingale.

**Definition 3.3 (Predictable Sequence)**

For a filtration  $(\mathcal{F}_n)_{n=0}^\infty$ , a sequence of random variables  $(H_n)_{n=1}^\infty$  is a **predictable sequence** if  $H_n$  is measurable with respect to  $\mathcal{F}_{n-1}$ .

It is immediate that predictable sequences are adapted, since the  $\sigma$ -algebras are increasing in a filtration, i.e.  $H_n$  is  $\mathcal{F}_{n-1}$ -measurable which implies  $H_n$  is  $\mathcal{F}_n$  measurable. Adapted sequences are not necessarily predictable.

**Example 3.4 (Gambling)**

At each round  $r$ , you bet  $\$x$ . With probability  $p$ , you win an additional  $\$x$  and with probability  $1 - p$ , you lose  $\$x$ . Let

1.  $X_n$  = not winning after  $n$  rounds, if you bet  $\$1$  each time.
2.  $H_n$  = amount of money you bet in round  $n + 1$ . So, makes sense to be  $\mathcal{F}_{n+1}$ -measurable. So, this is an example of a predictable sequence.
3. Now, our net earnings after  $n$  rounds of the game is

$$(H \cdot X)_n = \sum_{m=1}^n H_m(X_m - X_{m-1}) \quad (47)$$

where  $X_m - X_{m-1} \in \{\pm 1\}$ .

Now the strategy might be to double down if you lose, and keep betting  $\$1$  if you win, so

$$H_n = \begin{cases} 2H_{n-1} & \text{if } X_{n-1} - X_{n-2} = -1 \\ 1 & \text{if } X_{n-1} - X_{n-2} = \pm 1 \end{cases} \quad (48)$$

This betting strategy is called a martingale. Remember that if we had infinite money, we will always win. More generally, if  $(H \cdot X)_n$  is a submartingale, then  $\mathbb{E}[(H \cdot X)_n] \geq 0$ . So, we should consider cases which leads to  $(H \cdot X)_n$  being a submartingale.

**Theorem 3.3**

Let  $X_n$ ,  $n \geq 0$  be a supermartingale. If  $H_n \geq 0$  is predictable and each  $H_n$  is bounded (pointwise), then  $(H \cdot X)_n$  is a super-martingale.

*Proof.* We have

$$\mathbb{E}[(H \cdot X)_{n+1} \mid \mathcal{F}_n] = (H \cdot X)_n + \mathbb{E}[H_{n+1}(X_{n+1} - X_n) \mid \mathcal{F}_n] \quad (49)$$

$$= (H \cdot X)_n + H_{n+1} \underbrace{\mathbb{E}[X_{n+1} - X_n \mid \mathcal{F}_n]}_{\leq 0} \quad (50)$$

$$\leq (H \cdot X)_n \quad (51)$$

**Definition 3.4 (Stopping Time)**

We call a natural-valued random variable  $N$  a **stopping time** if

$$\{N = n\} = N^{-1}(n) \in \mathcal{F}_n, \quad (\mathcal{F}_k \subseteq \mathcal{F}_n \forall k \leq n \implies \{N = k\} \in \mathcal{F}_n) \quad (52)$$

This is abstract, but just think of it as a strategy of stopping a betting game.

**Theorem 3.4**

If  $N$  is a stopping time, and  $X_n$  is a supermartingale, then  $X_{N \wedge n}$  (where  $N \wedge n$ ) is a supermartingale. This is a stopped version.

*Proof.* We can write

$$X_{N \cap n} = \left( \sum_{m=1}^n H_m (X_m - X_{m-1}) \right) + X_0 \quad (53)$$

where  $H_m = \mathbb{1}(N \geq m)$ . Check that it is predictable, and then conclude by theorem 1.

**Example 3.5**

Suppose  $(X_n)_{n \geq 0}$  is a submartingale. Let  $a < b$ ,  $N_0 = -1$ . For  $k \geq 1$ , let

1.  $N_{2k-1} = \inf\{m > N_{2k-2} : X_m \leq a\}$ , i.e. when are you going below  $a$ ?

2.  $N_{2k} = \inf\{m > N_{2k-1} : X_m \geq b\}$ , i.e. when are you going above  $b$ ?

We claim that  $N_j$  are stopping times. We call  $U_n = \sup\{k : N_{2k} \leq n\}$  as the number of upcrossings completed by time  $n$ . Next, we have the upcrossing inequality, which is used to prove the martingale convergence theorem.

**Theorem 3.5 (Upcrossing Inequality)**

If  $(X_m)_{m \geq 0}$  is a submartingale, then

$$(b - a)\mathbb{E}[U_n] \leq \mathbb{E}[(X_n - a)^+] - \mathbb{E}[(X_0 - a)^+] \quad (54)$$

The martingale convergence theorem is like an analogous version of the Bolzano-Weierstrass theorem—that a monotone sequence bounded above converges.